

D031

Advancing Evidence-Based Innovation in Nursing Practice

Armored Drill

8 concepts · 12 items · 4 transfer items

What this is. Every concept in this course's public description, re-asked with more disguise than a practice test uses. Each stem carries facts that do not decide it. Every wrong answer is a named trap.

Source. Scope from the course's public description in the WGU Institutional Catalog (September 2026); items written from the field's standard body of knowledge. No WGU assessment item, course material or proprietary content is used or reproduced. Not affiliated with or endorsed by Western Governors University.

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How to run it

Paper first. Pencil. One pass, no notes, a timer if you want the pressure.

1. **Before you read the options**, fill the strip under the item: the target word (what is actually being asked), and the one fact in the stem that decides it.
2. **Principle before arithmetic**. Name the rule, then compute.
3. **Camouflage**. Most of every stem does not matter. Ask: which single fact decides this?
4. **Inverse pairs are the bait**. If two options are opposites, one of them is the trap built for you.
5. **Mark sure, unsure or guess** honestly. Sure-and-wrong is the most valuable miss on the page — it is a wrong mental model.
6. **Absolutes** (always, never, guarantee) are usually wrong. And recognizing an answer is not the same as owning it.
7. Items with a **heavy black bar** on the left are **transfer items**: the same principle, reversed or solved for a different variable.
8. Stop at the STOP page. Then grade, fill the triage grid, and tally the traps you fell for. The biggest tally is what you drill next.

The strip, filled in, looks like this:

Target word _____	The one fact that decides _____	Answer _____	[] sure [] unsure [] guess
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Items

1.

A nurse innovator at a hospital with a new rooftop helipad is judging evidence on chlorhexidine bathing to prevent bloodstream infections. She finds an expert opinion piece by a famous infection-control leader, a single-site cohort study, one RCT, and a meta-analysis of eight RCTs. Which is strongest?

- A. The single RCT.
- B. The meta-analysis of eight RCTs.
- C. The cohort study, because it reflects real-world practice.
- D. The expert opinion, because the author is famous.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

2. [TRANSFER — reversed stem / different variable]

Which source ranks lowest for judging whether an intervention works?

- A. A cohort study.
- B. A systematic review of RCTs.
- C. An editorial expressing expert opinion.
- D. A randomized controlled trial.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

3.

A team at a clinic with a new electronic check-in drafts this question: “In adults with heart failure, does telemonitoring improve outcomes?” The clinic serves 3,000 patients. What is most clearly missing?

- A. The intervention.
- B. The clinic's patient count.
- C. A comparison, a specific outcome, and a time frame.
- D. The population.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

4. [TRANSFER — reversed stem / different variable]

In the question “In older adults after hip surgery, does early walking compared with bed rest reduce pneumonia within 7 days?” what is the O?

- A.** Pneumonia.
- B.** Bed rest.
- C.** Within 7 days.
- D.** Early walking.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

5.

A unit with a new family lounge wants to cut call-light response times by testing a rounding change for four weeks, with results used only on that unit. What kind of project is this?

- A.** Research, because every project must go through full ethics review.
- B.** Research, because data are collected.
- C.** Evidence-based practice implementation of a new guideline.
- D.** Quality improvement.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

6.

A trial with 40,000 patients found a new wound dressing shortened healing by 0.3 days ($p = 0.001$). The dressing is sold by a large supplier and costs three times more than standard dressings. What is the best interpretation?

- A.** Statistically significant but likely not clinically meaningful given the small effect and higher cost.
- B.** Clinically important, because p is very small.
- C.** Unreliable, because the supplier is large.
- D.** Not significant, because 0.3 days is small.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

7. [TRANSFER — reversed stem / different variable]

A study of 30 patients found a 25% drop in falls with a new program ($p = 0.08$, $\alpha 0.05$). What is the best interpretation?

- A.** Clinically meaningless because p is above 0.05.
- B.** Not statistically significant; a large, possibly meaningful effect warrants a larger study.
- C.** Proves the program does not work.
- D.** Statistically significant because the effect is large.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

8.

In a trial of a discharge-coaching program run by a nurse with 20 years of experience, 30-day readmissions were 20% in the usual-care group and 15% in the coached group. What is the number needed to treat to prevent one readmission?

- A.** 20
- B.** 7
- C.** 5
- D.** 4

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

9. [TRANSFER — reversed stem / different variable]

An intervention has an NNT of 25. What is the absolute risk reduction?

- A.** 96%
- B.** 25%
- C.** 4%
- D.** 0.25%

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

10.

A nurse-led team at a hospital with a strong basketball-sponsorship program is excited about buying smart glasses for handoffs. No one has observed handoffs or asked nurses what slows them down. What should the team do first?

- A.** Roll out smart glasses to all units.
- B.** Buy a small number of smart glasses to test.
- C.** Pick the technology with the best reviews.
- D.** Observe handoffs and interview nurses to understand their needs.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

11.

A pressure-injury bundle cut injuries by half on one unit. The unit's manager is well liked and the unit recently moved floors. Eighteen months later no other unit uses it. Which step would most help it spread?

- A.** Move other units to new floors.
- B.** Mandate it everywhere at once.
- C.** Wait, since good results always spread on their own.
- D.** Recruit respected early adopters on other units to trial it and share visible results.

Target word _____	The one fact that decides _____	Answer ____	☐ sure ☐ unsure ☐ guess
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12.

Nurses from five hospitals with different electronic records meet monthly by choice to share what they are learning about early mobility, trying ideas and reporting back. The group has 40 members. What best describes this group?

- A.** A focus group.
- B.** A community of practice.
- C.** A formal governance committee.
- D.** A project team.

Target word _____	The one fact that decides _____	Answer ____	☐ sure ☐ unsure ☐ guess
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STOP

Answers and dissection follow. Finish every item and every strip first.

Key and dissection

Struck-through text is the camouflage: facts that do not decide the item.

1. Answer B Concept 1

A nurse innovator ~~at a hospital with a new rooftop helipad~~ is judging evidence on chlorhexidine bathing to prevent bloodstream infections. She finds an expert opinion piece ~~by a famous infection control leader~~, a single-site cohort study, one RCT, and a meta-analysis of eight RCTs. Which is strongest?

Target word: strongest evidence **Decides:** Meta-analysis of RCTs tops the hierarchy.

A	Dropped element	Strong, but one trial ranks below a meta-analysis of many.
B	Correct	Pools multiple randomized trials.
C	Adjacent category	Observational designs rank below RCTs for intervention effects.
D	Camouflage	Fame does not raise the level of evidence.

2. Answer C Concept 1 • transfer

Which source ranks lowest for judging whether an intervention works?

Target word: lowest level **Decides:** Expert opinion is at the base.

A	Adjacent category	Observational, but above expert opinion.
B	Inverted operation	The highest level.
C	Correct	Opinion without systematic data is the lowest level.
D	Confusion set	High level, below systematic reviews.

3. Answer C Concept 2

A team ~~at a clinic with a new electronic check-in~~ drafts this question: “In adults with heart failure, does telemonitoring improve outcomes?” ~~The clinic serves 3,000 patients.~~ What is most clearly missing?

Target word: missing PICOT elements **Decides:** No comparison, vague outcome, no time frame.

A	Confusion set	Telemonitoring is the intervention.
B	Camouflage	Size is not a PICOT element.
C	Correct	PICOT needs C, a measurable O, and T.
D	Inverted operation	Adults with heart failure is the population.

4. Answer A Concept 2 • transfer

In the question “In older adults after hip surgery, does early walking compared with bed rest reduce pneumonia within 7 days?” what is the O?

Target word: identify O **Decides:** The outcome measured.

A	Correct	The outcome is pneumonia.
B	Inverted operation	That is the comparison.
C	Adjacent category	That is the time frame.
D	Confusion set	That is the intervention.

5. Answer D Concept 3

A unit ~~with a new family lounge~~ wants to cut call-light response times by testing a rounding change for four weeks, with results used only on that unit. What kind of project is this?

Target word: project type **Decides:** Local process improvement with rapid cycles.

A	Absolute language	QI often uses a separate review path.
B	Confusion set	Collecting data does not make it research.
C	Adjacent category	No external evidence is being translated.
D	Correct	Improves a local process using rapid cycles.

6. Answer A Concept 4

A trial with 40,000 patients found a new wound dressing shortened healing by 0.3 days ($p = 0.001$). ~~The dressing is sold by a large supplier~~ and costs three times more than standard dressings. What is the best interpretation?

Target word: significance interpretation **Decides:** Tiny effect, large sample, high cost.

A	Correct	A low p-value does not guarantee a meaningful effect.
B	Confusion set	Confuses statistical with clinical significance.
C	Camouflage	The supplier does not decide the interpretation.
D	Inverted operation	The p-value shows statistical significance.

7. Answer B Concept 4 · transfer

A study of 30 patients found a 25% drop in falls with a new program ($p = 0.08$, $\alpha 0.05$). What is the best interpretation?

Target word: interpret non-significant large effect **Decides:** p above alpha with a small sample.

A	Inverted operation	Clinical meaning is judged by effect size, not p alone.
B	Correct	The effect may be real but the sample is too small to rule out chance.
C	Absolute language	Failure to reach significance is not proof of no effect.
D	Confusion set	Effect size does not set significance.

8. Answer A Concept 5

In a trial of a discharge-coaching program ~~run by a nurse with 20 years of experience~~, 30-day readmissions were 20% in the usual-care group and 15% in the coached group. What is the number needed to treat to prevent one readmission?

Target word: NNT **Decides:** ARR = 5%.

A	Correct	$1 / 0.05 = 20$.
B	Confusion set	$1 / 0.15$, using the coached group's rate.
C	Wrong stem number	The absolute risk reduction as a whole number.
D	Inverted operation	Uses the 25% relative risk reduction ($1 / 0.25$).

9. Answer C Concept 5 · transfer

An intervention has an NNT of 25. What is the absolute risk reduction?

Target word: solve for ARR **Decides:** $ARR = 1 / NNT$.

A	Inverted operation	Subtracts 4% from 100%.
B	Wrong stem number	The NNT read as a percent.
C	Correct	$1 / 25 = 0.04$.
D	Confusion set	Misplaces the decimal.

10. Answer D Concept 6

A nurse-led team at a hospital ~~with a strong basketball sponsorship program~~ is excited about buying smart glasses for handoffs. No one has observed handoffs or asked nurses what slows them down. What should the team do first?

Target word: first design thinking step **Decides:** Empathize with users.

A	Dropped element	Skips empathy, definition, and testing.
B	Inverted operation	Prototypes a solution before defining the problem.
C	Adjacent category	Choosing a tool is not understanding user needs.
D	Correct	Design thinking starts with empathy.

11. Answer D Concept 7

A pressure-injury bundle cut injuries by half on one unit. ~~The unit's manager is well liked and the unit recently moved floors.~~ Eighteen months later no other unit uses it. Which step would most help it spread?

Target word: spread strategy **Decides:** Opinion leaders, observability, trialability.

A	Camouflage	The move is unrelated.
B	Inverted operation	Skips trialability and adoption support.
C	Absolute language	Diffusion needs active support.
D	Correct	Opinion leaders and observable results drive diffusion.

12. Answer B Concept 8

Nurses from five hospitals with different electronic records meet monthly by choice to share what they are learning about early mobility, trying ideas and reporting back. ~~The group has 40 members.~~ What best describes this group?

Target word: group type **Decides:** Voluntary, shared domain, ongoing learning.

A	Inverted operation	A one-time research data collection method.
B	Correct	Voluntary, ongoing shared learning in a common domain.
C	Adjacent category	Committees hold decision rights in a hierarchy.
D	Confusion set	Project teams have assigned members and a fixed deliverable.

Triage grid

Write each item number in the box it belongs to.

	Right	Wrong
Sure	Owned.	Concept inversion — top priority. A wrong mental model; it survives rewording.
Unsure / guess	Recognition, not mastery. Re-drill with the transfer.	Not learned yet.

Trap tally

For every miss, find the letter you chose in the key and tally its trap. The biggest number is the pattern to drill.

Trap	What it does	In this drill	Your misses
Dropped element	Leaves out a required term or step.	2	
Wrong stem number	A figure copied from the prompt that is not the answer.	2	
Adjacent category	The right idea in the wrong bucket.	6	
Inverted operation	Direction flipped: add/subtract, cause/effect, higher/lower.	10	
Absolute language	Always, never, guarantee — where judgment applies.	3	
Confusion set	Every option is real; the test is exact role-matching.	9	
Camouflage	Answers a stem fact that was never the question.	4	

Anchor rows — the concept list

One row per concept: the rule to lock in, the trap that catches people, and the tell in the question that signals it.

#	Concept	Rule	Trap	Tell
1	Evidence hierarchy	For questions about intervention effects, systematic reviews and meta-analyses of randomized controlled trials rank highest, then single RCTs, then quasi-experimental, observational, and finally expert opinion.	Ranking evidence by the author's fame or the journal's prestige.	Several sources on the same intervention question.
2	PICOT questions	Population, Intervention, Comparison, Outcome, Time frame; each element must be specific and measurable.	Leaving out the comparison or naming a vague outcome.	A draft clinical question missing a piece.
3	Research vs. EBP vs. QI	Research generates new generalizable knowledge; evidence-based practice applies existing best evidence; quality improvement improves a local process with rapid cycles.	Calling a local process improvement project research.	Whether the goal is new knowledge, applying evidence, or improving a local process.
4	Statistical vs. clinical significance	A p-value below the set alpha shows a result is unlikely due to chance; clinical significance asks whether the effect size matters to patients.	Assuming a significant p-value means a meaningful benefit.	A tiny effect with $p < 0.05$ in a very large sample.
5	Absolute risk reduction and NNT	$ARR = \text{control event rate} - \text{intervention event rate}$; $NNT = 1 / ARR$.	Using relative risk reduction to compute NNT.	Event rates for two groups.
6	Design thinking	Innovation starts with empathy for users' real needs, then defines the problem, generates ideas, prototypes, and tests.	Jumping to a technology solution before understanding the user's problem.	A team excited about a tool before talking to users.
7	Diffusion of innovation	Spread depends on relative advantage, compatibility, simplicity, trialability, and observability; early adopters and opinion leaders carry an innovation to the majority.	Expecting a proven innovation to spread on its own.	A good idea not spreading beyond one unit.
8	Communities of practice	A community of practice is a voluntary group sharing a domain and learning together over time through shared practice; it is not a hierarchy or a one-time project team.	Treating a formal committee with assigned tasks as a community of practice.	A group formed to share learning across units vs. to deliver a fixed product.

Course scope (public description): Advancing Evidence-Based Innovation in Nursing Practice introduces students to the dynamic union of healthcare innovation and evidence. Core competencies and behaviors required to be a nurse innovator are discussed. Strategies for measuring innovation at various system levels are presented, as well as techniques for synthesizing and disseminating evidence to advance innovation in healthcare. The skills needed to appraise the quality of diverse sources of evidence are presented within the framework of

evidence-based practice. This course focuses on identifying new and emerging sources of evidence that can inform, translate, and scale the complexity of leading innovation in healthcare organizations. Students will experience building communities of practice for collaboratively developing innovative practices and policies designed to improve the health of populations and enhance the patient experience of care.